

Metadata Exclusion in Retrieval Services: Detection and Repair Evidence

Abstract—Retrieval services may retain metadata influence in stored vectors after an embedding policy excludes the field. Input-policy checks alone cannot establish whether served scores retain this influence or whether the available evidence suffices to accept repair. We address this gap with an audit protocol that separates fault detection from repair acceptance. Controlled field histories and serving-state checks define paired target-score comparisons; valid score intervals yield supported effect, no detectable effect, or insufficient evidence and guide further acquisition. We reproduce a metadata-update regression in official LlamaIndex releases. In a synthetic medical-record pipeline, all 48 effects persist under a stale reader; correcting the reader removes them but leaves top-ten evidence unresolved. Bound target scoring resolves all comparisons with 98.91 percent fewer response-body bytes than full-depth retrieval at the same request budget. The results concern finite score obligations for the declared histories and queries under trusted acquisition and state binding, without establishing deletion or answer-level privacy.

Index Terms—Metadata exclusion, retrieval systems, privacy auditing, evidence provenance.

I. INTRODUCTION

Retrieval-augmented generation (RAG) connects language models to external collections whose contents can change independently of model training [1]. Dense retrieval represents document content through learned embeddings [2]; its inputs may also include metadata governed by a service’s indexing policy. Recent work shows that RAG systems can expose private records and reveal database membership through their outputs [3], [4], [5]. These findings motivate a related question about changes in permitted use: when a service excludes a metadata field from its embedding policy but retains the document, what evidence establishes that the field no longer influences the served representation?

The distinction between an input policy and its downstream realization is fundamental to this question. An updated encoder configuration constrains future computation, whereas retrieval evaluates representations already stored and selected by the serving reader. Historical vectors can therefore preserve an excluded field’s influence after a policy change. Replacing those vectors addresses persistence only if retrieval uses the replacement state. For medical records, for example, a service may exclude a free-text comment while retaining the record and permitted clinical content. Its immediate verification

problem concerns the field’s influence on specified target scores, rather than document membership or complete data deletion. This score obligation must be chosen explicitly by the policy owner; it is not itself a general privacy guarantee.

Existing research provides complementary verification objectives. Work on auditable machine unlearning asks what observations can substantiate removal claims [6], [7]; trusted-execution and cryptographic approaches bind claims to prescribed computations [8], [9]. Retrieval-specific unlearning changes query–document associations or knowledge-base contents to suppress designated information [10], [11]. Personal-data audits instead assess unauthorized participation or observable disclosure [12], [13]. These objectives do not directly determine whether a fixed document’s native retrieval score is invariant to one field’s history. Here the encoder and document membership remain fixed, while the evidence must distinguish a compliant change in derived state from a still-dependent representation.

A further difficulty arises from the limited observability of top- k interfaces. Detecting an effect can require less evidence than establishing its absence. A target visible in one field world may exceed a valid upper bound on its omitted counterpart, yielding a witness of dependence. After repair, however, both target scores may fall outside the returned set. The disappearance of the witness then leaves the comparison unresolved, even when the underlying scores are identical. This is an evidence problem created by censoring: observations adequate to reject an obligation need not be adequate to accept its repair. We therefore ask *when do paired retrieval observations suffice for detection and acceptance, and which additional evidence resolves the remaining uncertainty?*

We study a finite obligation on the same target’s score under controlled field histories. Stable identities, permitted inputs and queries define the contrast; trusted acquisition and serving-state checks establish comparability. Visible scores, documented numerical uncertainty and valid omission bounds define intervals for the score difference. Classifying the entire interval yields supported effect, no detectable effect, or insufficient evidence. The approach makes the boundary of each conclusion explicit and identifies when greater depth, precision or trusted

target scoring is required. Complete same-state scores remain sufficient and subsume valid interval judgments; our contribution concerns the interpretation and acquisition of incomplete evidence.

We connect these evidence requirements to native update and reader behavior through a reproduced upstream regression and controlled interventions. We make three contributions:

- 1) To isolate an excluded field’s influence from other changes, we propose **a paired-history audit protocol** that controls permitted inputs and target identity while binding native-score comparisons through trusted serving-state checks.
- 2) To determine what censored top- k observations can establish, we formulate **an interval evidence framework** that separates fault detection from repair acceptance and identifies sufficient additional evidence under explicit applicability conditions.
- 3) To explain why successful repair can remain unverifiable, we conduct **a mechanism-focused evaluation** of an upstream update regression and controlled reader interventions. Score and rank distributions expose the loss of visibility; matched acquisition experiments test which channels resolve the resulting uncertainty.

II. RELATED WORK

a) Privacy and integrity of retrieval-augmented generation: RAG privacy studies examine disclosure of retrieved records [3], [14] and mitigation through synthetic retrieval data [15]. Membership audits use shadow systems, similarity signals or answer behavior to infer participation in the retrieval corpus [4], [5], [12], [13], [16]. Prompt-injection benchmarks and corpus-poisoning attacks address adversarial control of generated outputs [17], [18]. These studies motivate verification at the retrieval boundary, but their observables differ from ours: we fix corpus membership and test a controlled field’s effect on the same target’s native score. Neither a membership verdict nor an answer-level attack directly establishes this score invariant.

b) What constitutes verifiable removal: Unlearning research distinguishes an intended removal from evidence that the prescribed change occurred. Thudi et al. analyze the dependence of forgetting metrics on training dynamics and the necessity of auditable definitions [6], [19]; Zhang et al. demonstrate fragility in verification and study certified removal from deep networks [20], [21]. Athena and invisible-trigger approaches use statistical or backdoor evidence [22], [23]. Proof of Unlearning and cryptographic verification bind claims to execution under their respective trusted-hardware and proof assumptions [8], [9]. Wang et al. measure sample-level unlearning completeness beyond binary inclusion

tests [24], while a recent survey organizes the reliability of verification criteria [7]. Our encoder remains fixed: the relevant state is the retrieval representation and its serving path. Saved hashes establish consistency of supplied evidence, without providing those execution guarantees.

c) Retrieval-specific forgetting and residual state: CABOOSE removes user interactions from neighborhood recommendation models [25]; CuRD learns to suppress query–document associations [10]; RAG unlearning examines behavioral forgetting through knowledge-base changes [11]. RESTOR evaluates forgetting together with recovery of the knowledge state that would have existed without the removed data [26]. Recent preprints expose related persistence mechanisms: Ghost Vectors studies soft-deleted HNSW embeddings and Ghost Echoes studies retrieval-context drift after deletion [27], [28]. Those interventions change membership or model behavior. Our fixed-membership contrast instead separates field influence, update completion and the evidence needed to accept a repaired score obligation.

d) Statistical auditing and relational evidence: Privacy audits obtain access-dependent lower bounds from controlled interventions and randomized output distributions [29], [30]. We share their need to distinguish a detected violation from an absence of detection, but do not estimate a differential-privacy parameter or an output distribution. Our tolerance is a score guard, and soundness is conditional on deterministic interval containment. MeTMaP applies metamorphic relations to vector-matching behavior [31]; our relation concerns the same target’s absolute score, so changed ranks or reference gaps cannot substitute for the specified contrast.

e) Retrieval quality and score-bound validity: Representation benchmarks assess embedding quality across tasks [32]; INSTRUCTOR conditions embeddings on task instructions [33], and BGE-M3 supports multilingual retrieval [2]. Faiss provides the native similarity-search infrastructure used in our analysis [34]. These results concern retrieval quality and implementation capabilities. An audit additionally needs a bound on unobserved target scores under its exact state and precision assumptions. Our supplementary study applies established cosine triangle bounds [35] and multi-reference simplex geometry [36]; nSimplex embeddings provide related geometric context [37]. A recent identified-set preprint studies top- k censoring for model extraction [38], with conditional token distributions constrained by returned top- k logits as its estimand. We use existing geometry to constrain a shared stored vector and retain its zero decision gain; we do not claim new pivot geometry or optimizer optimality.

III. TRUSTED-HOST AUDIT MODEL

a) *Objective, faults, and auditor:* The objective is to avoid accepting a field-exclusion claim when its score obligation fails. We evaluate configuration and migration faults on a trusted host: retained vectors, wrong write destinations and mixed readers. An authorized auditor controls variants/queries and obtains administrator-assisted snapshots and startup access. Acquisition and host integrity are assumptions (Supplementary Figure 5 and Table III).

b) *Evaluated trust model:* Receipts identify write destinations, not serving stores. Descriptors require trusted acquisition; hashes check pinned bytes without authenticating execution. Section IV-B’s client binds channels inside this local boundary. A host controlling the evidence producer is excluded.

c) *Why audit inside this boundary:* An internal team can quiesce writes and inspect the reader while distrusting migration completion. External review needs equivalent trusted access; institutional status alone supplies none. Attestation such as Proof of Unlearning [8] addresses execution trust under its own protocol; it does not supply our histories or score obligation. Selective fabrication and hidden replicas remain outside our guarantee.

d) *Outside the model: a dishonest host:* Let T be everything a transcript-only auditor can observe. Compare a compliant production service with a violating service whose operator answers every audit interaction using an identical compliant copy, including artifacts and labels. Without an independent binding to production, these worlds induce the same distribution of T , even for adaptive randomized audits. Thus every decision rule has the same acceptance probability in both worlds. Self-reported hashes or signatures cannot distinguish them: the dishonest operator can report authentic properties of the compliant copy. Independent acquisition or an appropriate attestation/proof mechanism is needed to bind the audited state.

Ordinary outsider access lacks these controls; membership stays fixed, and no access-control bypass is assumed. Under Section IV-B’s conditions, supported effects falsify the score obligation. Score nulls establish neither nonprocessing, deletion nor anonymity. Our synthetic DICOM and public-text studies provide no hospital or commercial deployment validation.

IV. METHOD: AUDITING A TARGET’S SCORE RESPONSE

Figure 1 summarizes the audit: validate the controlled contrast, construct comparable target-score intervals, and classify their difference. Repair acceptance requires sufficient evidence for every declared comparison.

a Controlled field contrast

Receipts + stable IDs; fixed body, membership and insertion order; only prescribed fields vary.

b Comparable field intervention

Stable query, encoder, scorer and state; target-local encoding or one-record edit for single-record attribution.

c Valid target-score intervals

Use documented score/error bounds. Missing: exact top-k or certified bound; else use an uninformative interval.

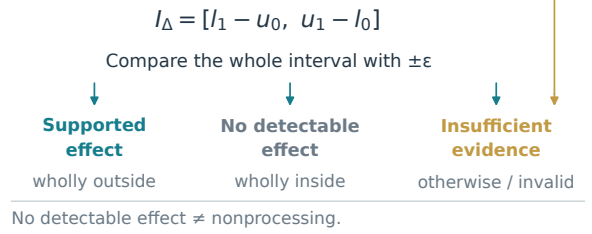


Fig. 1. Target-score audit flow. Unverified pairing, intervention or score comparability yields insufficient evidence. Valid intervals give one of three verdicts; missing targets without certified bounds receive uninformative intervals. An unresolved comparison requires stronger authorized evidence or continued abstention. Repair acceptance requires all declared comparisons to have no detectable effect under the checked conditions and recorded trust assumptions.

A. Estimand and controlled contrast

We define the signed change in the cosine score of the *same logical target* a between two field phases:

$$\Delta_a = s_1(q, a) - s_0(q, a). \quad (1)$$

We test $|\Delta_a|$ in (1) against $\epsilon = 10^{-6}$, fixed before evaluation as an operational guard, not a universal numerical or privacy bound. Ranks and target-reference gaps do not identify it: competitors or the reference may change while $s_j(q, a)$ remains fixed.

The finite set $\mathcal{A}$ declares query-target comparisons, paired histories, state S , and the obligation $|\Delta_a| \leq \epsilon$ throughout. Acceptance requires every difference interval within $[-\epsilon, \epsilon]$ under checked conditions and recorded trust assumptions; it covers no untested values, queries, or versions.

This is a proposed diagnostic for derived-state migration, suitable only when the policy owner has chosen a finite score obligation. The service owner defines permitted representations and query coverage; the policy owner decides whether score invariance is appropriate and fixes its tolerance before measurement. We have no

evidence of organizational adoption or validation of our experimental 10^{-6} guard as a privacy policy. For an obligation about ranks, returned payloads or generated answers, those outputs must be audited directly; the score diagnostic cannot replace that audit.

Supplementary Figure 6 defines attribution cohorts. Controls diagnose proxy rules; our target-score rule needs no reference score.

A migration audit compares both field worlds at the *same* stage. Comparing stages within one world can miss retained vectors that preserve both scores and dependence.

B. Applicability conditions

Table I specifies attribution conditions. Unverified acquisition, intervention, or comparability leaves field attribution insufficient even with observed score differences.

Simultaneous edits measure the declared joint field intervention. Single-record attribution requires target-local encoding or individual interventions, beyond document-local scoring and stable queries. Batch-induced null gaps are not universal error bounds; corpus-normalized, stateful, or reranked systems require further controls.

a) Acquisition and an existing API: Public checks validate identities, responses and verdicts; operator checks compare inventories, vectors, queries, writes, readers and complete-score prefixes. Trusted acquisition must bind these to the intended epoch, establish paired histories, and exclude unrecorded writers/replicas. Hashes do not attest those premises.

Pin versions and eligibility, snapshot or quiesce updates, obtain score/error contracts, and preflight sufficient channels (Section IV-D). Log restoration, encoding and writes; vector reuse requires locality. Without reconstructible history, clean-index tests concern prospective configuration only.

b) Implemented cooperative binding: Each world/version uses a separate process with one captured Haystack retriever/store and frozen queries. Trusted requests intersect target IDs with ordinary eligibility under document-local scoring. The synchronous epoch has no write/reload route.

A descriptor covers data/vectors, identities, scorer source/settings, queries, eligibility and precision, rechecked around requests. Clients validate epoch, world/version, query and identities. Channels share a descriptor within each world; cross-world checks enforce declared invariants. Binding covers only the inspected process (Section V-A).

c) Reusing assurances across updates: Reuse requires unchanged encoder/tokenizer, queries, backend, scorer, precision, filters and search semantics. Recheck

identity, membership, field selection, writes and reader binding on migration; fingerprints cannot transfer acceptance across versions.

d) Restricted contracts: Answer-only responses cannot supply target-score intervals. Dynamic corpora require matched snapshots or a justified drift model; our numerical guard does not compensate for uncontrolled state.

For known nearest rounding step h , a score r admits $[r - h/2, r + h/2]$ with outward numerical endpoints. Exact-search omissions have upper bound $c + h/2$ for rounded cutoff c . Scoring error further widens intervals. Our prospectively specified saved-response projections preserve retrieval order; they do not validate unknown API contracts.

An ANN cutoff supplies no omission certificate. An unseen cosine score's $[-1, 1]$ range cannot force an effect with another valid cosine score. Comparable visible scores remain usable; omitted targets need certified bounds. We supply no generic ANN or dynamic-corpus bound.

C. Intervals and three verdicts

For phase j , a visible unrounded score gives $I_j(a) = [s_j, s_j]$. Absence from verified exact top- k gives $I_j(a) = (-\infty, c_j]$, where c_j is the true returned cutoff. The unbounded lower endpoint is conservative; known score ranges may tighten it. Rounded or certified approximate scores use their documented intervals instead. If $I_j = [l_j, u_j]$, then

$$I_\Delta = [l_1 - u_0, u_1 - l_0]. \quad (2)$$

For (2), support requires the interval wholly above ε or below $-\varepsilon$; no detectable effect requires it inside $[-\varepsilon, \varepsilon]$. Otherwise abstain, including two omissions. Reports distinguish failed applicability, invalid responses and overlap. The ANN heuristic deliberately tests an uncertified premise.

a) Proposition: conditional interval soundness: Fix the audit set, states and $\varepsilon \geq 0$. Assume valid pairing, intervention, comparable scores and trusted provenance, with every interval $I_j = [l_j, u_j]$ containing its actual score s_j . Acceptance of every declared comparison as no detectable effect establishes the finite obligation under these premises. Specifically, the rule in Algorithm 1 satisfies

$$\begin{aligned} \text{supported effect} &\implies |s_1 - s_0| > \varepsilon, \\ \text{no detectable effect} &\implies |s_1 - s_0| \leq \varepsilon. \end{aligned} \quad (3)$$

Proof. Interval containment gives $l_1 - u_0 \leq s_1 - s_0 \leq u_1 - l_0$. The support branch places this entire interval outside $[-\varepsilon, \varepsilon]$; the no-effect branch places it inside. Applying the latter implication to each comparison proves the acceptance claim. $\square$

TABLE I
AUDIT APPLICABILITY. THE CUTOFF CONDITION IS REQUIRED ONLY WHEN USING AN OMITTED TARGET’S SCORE BOUND. UNSUPPORTED CONDITIONS CANNOT BE REPAIRED BY TREATING ABSENCE AS A NEGATIVE RESULT.

Check	Required evidence	If unavailable
Pairing	Receipts and stable identity map link the same target across fixed membership, body, and order	Insufficient field-effect evidence
Intervention	The specified joint field intervention is isolated; other input and execution changes cannot affect the target score	Score association only; field attribution unresolved
Query and state	Identical query/encoder input, encoder/scorer version, candidate filters, and controlled execution state	Snapshot/control the state; otherwise insufficient attribution
Score scale	Comparable absolute scores, or a documented error/rounding interval on a common scale	Rank or answer changes do not identify absolute-score change
Censoring	Exact top-k over the declared corpus, or a certified upper bound for omitted targets	No cutoff certificate; use visible scores or an uninformative missing-target interval
Numerics	A fixed decision guard and assessed numerical uncertainty represented in the score intervals	Do not infer a field effect from an unbounded numerical perturbation

Equation (3) gives soundness, not completeness: a true effect may receive insufficient evidence. If insufficiency counts as non-detection, its rate is unbounded by the proposition. The result therefore does not assert zero Type-II error or a calibrated statistical test, and does not authenticate the premises or unobserved production state. The empirical questions are evidence coverage, misattribution, and acquisition costs; score verdicts leave processing paths unresolved.

b) Complete scores subsume interval judgments:

For the same finite obligation and trusted state, exact target scores determine $\Delta_a \in I_\Delta$. Every decisive interval verdict therefore follows from direct comparison of $|\Delta_a|$ with ε ; singleton intervals give identical rules. Complete reader/version checks plus these scores cannot be surpassed in decision coverage by valid intervals. Direct rescoring must preserve the production query, scorer, eligibility and state. A correct baseline with uncertain scores must honor their error contract too. This is an information limit for the score obligation, not a new adversarial guarantee.

D. Selecting precision and repair evidence

Owners fix tolerance before measurement; numerical uncertainty belongs in the interval. For two visible reported scores, use conservative intervals $I_j = [r_j - e_j, r_j + e_j]$, with $e_j \geq 0$ covering documented rounding and scoring error. Write $d = r_1 - r_0$ and $w = e_0 + e_1$. Without score-range clipping,

$$\begin{aligned} &\text{effect: } |d| - w > \varepsilon, \\ &\text{no detectable effect: } |d| + w \leq \varepsilon. \end{aligned} \quad (4)$$

Intermediate cases are insufficient; clipped or asymmetric intervals use endpoints. Preflight $w \leq \varepsilon$: otherwise equal reports alone cannot establish no effect. Nearest-rounding step h gives $w = h$; six decimals permit acceptance at our guard, four do not. Increase precision

Algorithm 1 Audit one target’s score response

Require: Target a , paired unrounded exact top- k responses R_0, R_1 , guard ε , applicability checks V (Table I)

Ensure: Effect verdict; processing path remains unresolved

```

1: if  $V$  fails or a response is invalid then
2:   return insufficient evidence
3: end if
4: for  $j \in \{0, 1\}$  do
5:   if  $a$  is visible in  $R_j$  then
6:      $[l_j, u_j] \leftarrow [s_j(a), s_j(a)]$ 
7:   else
8:      $[l_j, u_j] \leftarrow [-\infty, \text{cutoff}(R_j)]$ 
9:   end if
10: end for
11:  $[L, U] \leftarrow [l_1 - u_0, u_1 - l_0]$ 
12: if  $L > \varepsilon$  or  $U < -\varepsilon$  then
13:   return supported effect
14: else if  $-\varepsilon \leq L \leq U \leq \varepsilon$  then
15:   return no detectable effect
16: else
17:   return insufficient evidence
18: end if
```

or narrow validated error bounds; depth cannot repair precision. Supplementary Section C gives the measured boundary and tolerance sensitivity. Increase depth or request trusted target scores within the same pinned state; exhausted channels leave insufficiency.

E. Evaluation design and evidence channels

a) Connected migration case: Two field worlds each contain 192 synthetic DICOM records: 24 targets, 24 unchanged-null records, and 144 near-topic backgrounds. Orthanc receives uploads; Haystack fetches

simplified-tags JSON, converts TextValue to content, embeds with pinned MiniLM, and retrieves by exact cosine. The 24 queries cover twelve fictional medical themes. Only target comments differ; file metadata, UUIDs, pixels, permitted text, order and queries match. SOPInstanceUID pairs records. Source URLs and Haystack IDs differ across worlds, remain stable across migration stages, and are not encoded. This researcher-assembled chain uses real components, synthetic records and no LLM generation.

The obligation prohibits PatientComments from affecting served target scores. Each world includes comments and writes, then excludes comments with intentional duplicate-policy SKIP, then OVERWRITES the index. SKIP is a planted fault, not an upstream promise or undocumented behavior. Two targets and two null records per query yield 48 target and 48 null comparisons per stage. We compare configuration, encoder inputs, receipts, public top-ten intervals, and private full-192 scores; public decisions exclude private reconstruction. Saved protocols record formal inputs fixed before development, but two themes overlap: this is not an independently preregistered or theme-disjoint holdout.

Supplementary Section C records the fixed new-query, precision and near-threshold designs.

Section V-A gives the reader experiment. Supplementary Sections B and E record attribution controls and native-software protocols. Earlier indexing and competition measurements remain archived with their original negative results; they are not needed for the claims evaluated here.

V. EMPIRICAL ANALYSIS

The experiments examine three implications of the audit model: a policy change need not change served representations; removing an effect need not make repair observable; and retrieval accuracy need not establish a valid omission bound. We use native complete scores to evaluate these implications, while withholding them from the limited-evidence rules. Counts describe the fixed workloads and their dependent comparisons, rather than estimates of failure prevalence.

A. Why input exclusion can leave served influence intact

a) Update detection is part of the causal path:

We reproduce the metadata-update regression reported in LlamaIndex issue 17871 [39] using official core 0.12.27 and 0.12.28. The repair in pull request 18303 [40] is included in the 0.12.28 changelog [41]. In the affected release, document hashes omit metadata; changing metadata alone therefore fails to trigger replacement. A frozen two-document, one-query fixture removes the target's

Title while retaining its identity and body. The old release processes no update and preserves score 0.910012, whereas rebuilding the intended records gives 0.112141. The resulting residual is 0.797871. The fixed release processes the edit and matches the rebuild, with zero residual. Native hashing, storage and retrieval are used in both versions; the sixteen saved scores are independently reconstructed.

This contrast links the score residual to an omitted update, rather than to a changed query or scorer: dependency versions and inputs are matched, and the metadata-hash change explains why only the fixed release processes the edit. Both metadata and vectors remain in the affected store. The example therefore establishes a concrete failure mechanism, not successful metadata deletion with vector-only retention or the prevalence of such failures. The target also remains rank one, demonstrating that a large score residual need not change the top result.

b) *Changing the encoder input does not replace historical state:* The DICOM-Haystack experiment separates configuration, writes and serving state. Across two field worlds with fixed membership and permitted content, excluding PatientComments from current encoder inputs leaves all 48 target effects when duplicate-policy SKIP performs no replacement writes. Their absolute differences range from 0.336 to 0.718. Replacing all 384 vectors with OVERWRITE removes every complete-score effect. Saved-write replay reconstructs all 27,648 complete scores within 3.33×10^{-16} , with no query-vector drift. The mechanism is persistence of the old representations: the exclusion policy constrains newly computed inputs, but those representations do not affect retrieval until they replace the served vectors. SKIP is an intentional fault condition, distinct from the upstream regression.

c) *Successful writes do not identify the reader:* To isolate the remaining link, we restore the old and rebuilt vectors into separate native stores and direct the pipeline to the old store. All 384 writes to the rebuilt stores succeed and their identity inventories match, yet top-ten intervals still detect all 48 effects. Correcting only the reader binding removes every complete-score difference. Figure 2(a) shows the distribution: the old-reader median difference is 0.523328, with quartiles 0.449019 and 0.587361, whereas every corrected difference is exactly zero. Thus successful writes and matching record IDs establish replacement at a destination; they do not establish which representation participates in a served score. Reader inspection supplies that missing relationship and diagnoses this planted routing fault.

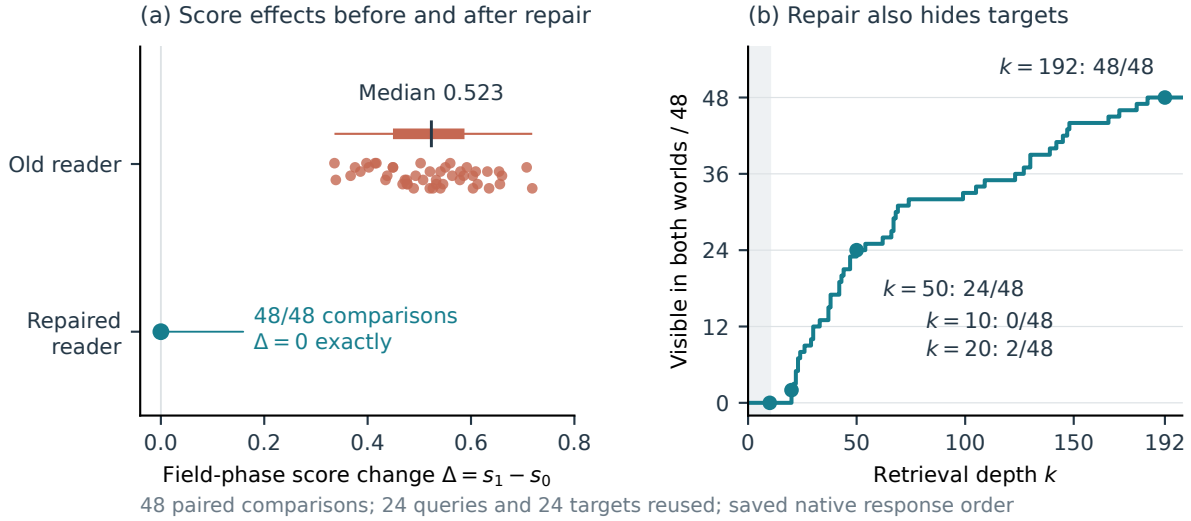


Fig. 2. Why removing an effect does not establish repair from top- k evidence. (a) Native score differences for 48 paired comparisons, each computed as world 1 minus world 0 within the indicated reader state. Thin/thick segments show the observed range/interquartile range; the tick marks the median. Vertical offsets only separate old-reader points; all repaired values coincide at zero. (b) Cumulative repaired-target coverage at the tested depths: 0, 2, 24 and 48 comparisons at $k = 10, 20, 50, 192$. Positions agree across worlds and retain native tie order. The comparisons reuse 24 queries and 24 targets; summaries are descriptive, not uncertainty estimates.

B. Why effective repair can reduce observable evidence

The contrast in Figure 2 explains the central asymmetry. Before repair, each target is visible in exactly one field world. A visible score above the other world’s valid cutoff can establish a difference even when the second score is omitted. For example, one target has score 0.630803 at rank two in one world and is absent from the other, whose top-ten cutoff is 0.490387. Its difference is therefore at least 0.140416, well above the guard. This is a rejection witness obtained without the missing score; it does not identify whether current processing or retained state caused the effect.

After repair, the same target has score 0.208110 and returned position 139 in both worlds. Its true difference is zero, but neither score is exposed at depth ten. Two omission upper bounds allow both equality and differences outside the guard; they cannot establish that the entire difference interval lies within tolerance. All 48 target comparisons undergo this loss of top-ten visibility. The change from 48 supported effects to 48 insufficient comparisons is consequently consistent with successful repair. Treating the disappearance of detections as acceptance would conflate a change in the system with a loss of observability.

Figure 2(b) connects this reasoning to the depth response. Repaired positions range from 20 to 184, with median 52 and quartiles 32.25 and 124. At the tested depths, both-world visibility resolves 0/48, 2/48, 24/48

and 48/48 comparisons. Full depth is the first tested sufficient setting; it is not the estimated minimum depth. The distribution explains why a modest increase from ten to twenty returns twice as many records but resolves only two comparisons: most repaired targets still lie below the cutoff. These post-hoc summaries retain native tie ordering and do not imply a corresponding change in semantic relevance or answer privacy.

The observations distinguish the score obligation from a proxy such as retrieval rank. The protocol attributes a difference in the same target’s absolute score under controlled field histories. Ranks determine which evidence is visible, but depend on competing scores and tie handling; they are not the audited outcome. This distinction also explains the supporting attribution experiment’s false positives from rank and reference-gap proxies (Supplementary Section B).

C. What additional evidence is sufficient for acceptance

a) Coverage depends on which scores are acquired:

Figure 3 compares evidence channels on the same repaired state. At a fixed budget of 48 calls, full-depth retrieval obtains 9,216 scores and 11,171,866 response-body bytes. Trusted retrieval filtered to the two declared target IDs per query obtains 96 scores and 121,694 bytes, resolving the same 48 comparisons with 98.91% fewer acquired bytes. The reduction follows from excluding unrelated records from the response, not from a stronger

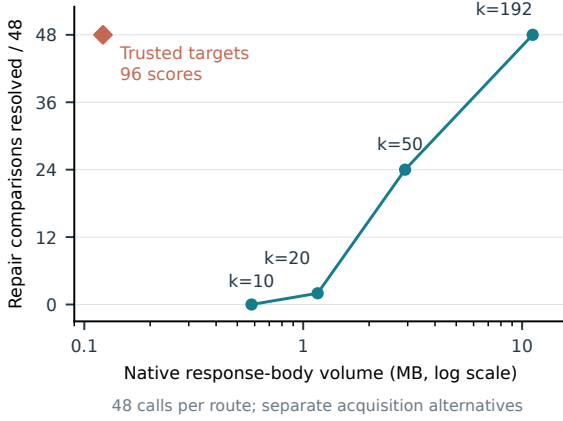


Fig. 3. Sufficient evidence and acquisition volume for the same 48 repaired comparisons. Each point uses 48 calls; connecting tested depths does not represent cumulative escalation cost. The target route adds score access and trusted state binding. Bytes include native response bodies and exclude headers and other audit costs.

decision rule. Both routes return native bodies including text and metadata; headers, state preparation and human effort are excluded. The subsequent score-only projection releases 10,684 bytes, which is a different quantity from acquisition.

The curve exposes why evidence should be selected for the obligation. Increasing depth acquires every higher-ranked competitor before a low-ranked target; direct target acquisition asks for the quantities in the score comparison. The latter is sufficient only if filtering preserves the query, scorer, target eligibility and served state. Our cooperative binding checks reject mixed-generation channels and inconsistent query/target identities. They check consistency within trusted startup, rather than authenticating an independently operated or adversarial host. Additional score access without that binding would not establish the declared obligation.

b) Equal access identifies the source of the gain:

Table II compares the protocol with reader inspection and direct rescoring under identical permissions. Both obtain the same target scores, detect 48/48 old-state effects and resolve 48/48 corrected-state comparisons. Their evidence volumes are identical and their local timing ranges overlap. Singleton intervals reduce algebraically to direct score subtraction; no detection or speed advantage is expected or observed. The useful distinction is therefore between sufficient and insufficient evidence channels, rather than between two equivalent rules once complete scores are available.

This baseline also limits the interpretation of the byte reduction. The native full-document comparison in Figure 3 measures what is acquired; the baseline’s bound responses contain descriptors and target ID/score

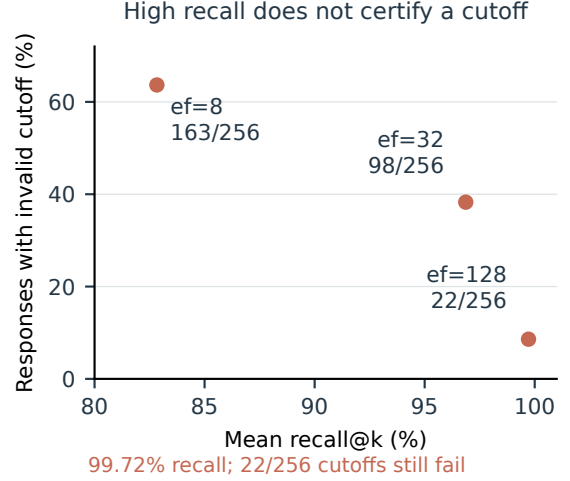


Fig. 4. Retrieval accuracy and cutoff validity measure different properties. Each search setting pools 256 native HNSW responses across the fixed queries, depths and field states. A cutoff fails if an omitted score exceeds it by more than 10^{-6} . At 99.72% mean recall, 22/256 responses still fail this condition. These failures invalidate a bound premise; they are not counts of incorrect final audit labels.

projections, so their byte totals describe a different response format. Neither comparison establishes the cost of recovering historical states or negotiating access. The matched protocol and all timing ranges remain in Supplementary Section E. Excluding two development-overlapping themes leaves the same detection/acceptance asymmetry on forty comparisons, a sensitivity analysis rather than an independent holdout.

D. Why high recall does not certify omitted scores

The cutoff argument relies on a property stronger than retrieval accuracy: no omitted score may exceed the claimed upper bound. We test this premise using native FAISS HNSWFlat [34], [42] on 5,183 saved vectors encoded from SciFact [43]. Four field-policy states, sixteen queries, four depths and three fixed search settings yield 768 responses. Exhaustive native scoring supplies a separate oracle; all returned scores match it exactly. Figure 4 compares recall and cutoff failures on these saved responses.

Increasing efSearch raises mean recall from 82.83% to 99.72%, while reducing invalid cutoffs from 163/256 to 22/256. The residual failures arise because recall averages recovered neighbors, whereas an omission bound must cover every omitted candidate. Missing even one candidate above the returned cutoff defeats that bound. Accurate scores for returned neighbors cannot repair the missing coverage. High recall is therefore compatible

TABLE II

EQUAL-ACCESS BASELINE ON THE SAME FROZEN STORES. BOTH METHODS INSPECT THE READER/VERSION AND USE THE SAME TRUSTED NATIVE TARGET-SCORE ROUTE. T/N/I ARE EFFECT, NO DETECTABLE EFFECT, AND INSUFFICIENT COUNTS OUT OF 48. TIMES ARE MEDIANS OF SIX LOCAL BATCHES AFTER WARMUP, EXCLUDING SHARED STARTUP. ORDINARY-ONLY ROWS ARE PREVIOUSLY COLLECTED, WEAKER-ACCESS REFERENCES WITH NO NEW TIMING.

Method	State	T / N / I	Calls / scores	Body bytes	Median s
Inspection + direct rescore	Old	48 / 0 / 0	48 / 96	34,020	5.469
Inspection + intervals	Old	48 / 0 / 0	48 / 96	34,020	5.471
Inspection + direct rescore	Corrected	0 / 48 / 0	48 / 96	34,034	5.365
Inspection + intervals	Corrected	0 / 48 / 0	48 / 96	34,034	5.378
Ordinary top ten	Old	48 / 0 / 0	48 / 480	88,683	—
Ordinary top ten	Corrected	0 / 0 / 48	48 / 480	88,608	—

with an invalid certificate, including at the largest tested search setting.

The distinction matters at two stages of the inference. Substituting the ANN cutoff produces target-interval containment failures in twelve paired cells, but no observed incorrect decisive labels: an interval can exclude its true score without crossing the decision threshold. Exhaustive omission bounds resolve nine additional paired cells relative to uninformative missing-score intervals, but require privileged access; complete target scoring settles all 432 dependent cells. The scan verifies this fixed state and supplies no generic HNSW certificate. This result supports abstention when the necessary bound is unavailable, not a claim that approximate search always misclassifies an obligation.

a) Narrower intervals need not change a decision:

A supplementary cross-query study tests another route to stronger bounds using classical single- and multi-query geometry. Across 4,880 dependent cells, every method returns the same 1,957 effects, 1,943 no-effect verdicts and 980 insufficiencies despite 998 narrowed missing-score intervals. All unresolved difference intervals contain zero and have median width 2.91579. Thus the observed tightening does not move an entire interval into the tolerance band or beyond an effect threshold. This negative result reinforces a decision-level criterion for evidence quality: interval contraction alone is insufficient. The full design, baselines and adverse results remain in Supplementary Section F.

VI. DISCUSSION

a) Detection and acceptance have different evidence requirements: The experiments instantiate an asymmetry in the finite obligation. Under valid pairing, state and score premises, one supported effect is a rejection witness; acceptance requires a within-tolerance result for every declared comparison. Repair changes both the system and which target scores a restricted interface reveals. Consequently, a sound audit should specify a detection route and a sufficient acceptance

route before migration. If the latter is unavailable, the absence of further detections leaves acceptance unresolved. This principle is an operational consequence of interval containment and the quantification over the audit set, not an additional theorem or a stronger guarantee than complete scoring.

b) An advantage requires an actual difference in available evidence: Valid cutoff evidence is useful when one observed score exceeds the counterpart’s omission bound. Its acquisition advantage requires trusted histories, state binding and valid omission bounds to be available while equivalent target scores are not. If verifying the cutoff already requires complete scores, that advantage disappears. The equal-access baseline makes this condition explicit: reader inspection plus direct scoring resolves exactly the same obligations. A documented role restriction also need not enforce the required separation. Our source review finds that standard Milvus Search can combine filtering and vector output; denying its separate Query operation does not exclude these stronger routes. The review is retained with the artifact and is not a service deployment experiment.

The present ordinary-only experiments constrain the rule’s inputs while experimenters retain administrator access. They therefore establish the consequences of an evidence restriction, not an independently enforced deployment in which that restriction is necessary. An external evaluation should enumerate all permitted routes, including filtered scoring and vector export, and compare resolution and acquisition cost under equivalent state and score guarantees. Measuring history recovery and operator effort would further distinguish a useful acceptance procedure from an inexpensive response format.

c) The score obligation is narrower than an output policy: A score effect can leave the returned identities unchanged, as in the rank-one upstream example. Conversely, changes in competing scores can move a target whose own score is invariant. Ranked-output stability depends on all eligible candidates and tie handling; payload exclusion and answer disclosure require their own observables. The proposed audit is appropriate when the

policy owner requires a finite score-invariance condition as a diagnostic of a derived index. It cannot substitute for deletion, nonprocessing, anonymity or answer-level obligations, and the experiments do not validate 10^{-6} as an organizational privacy tolerance.

d) *Limits of inference and generalization:* The experiments use controlled field interventions, a fixed primary encoder and document-local scoring. The 48 principal query–target pairs reuse 24 queries and 24 targets; the 4,880 cross-query cells reuse 305 pairs over engines, policies and depths. Distribution summaries describe these workloads, not independent trials or prevalence estimates. The natural effects lie above the guard; separate interpolation and rounding studies probe numerical limits without estimating natural occurrence rates. A known rounding projection cannot validate an undocumented service contract, and high HNSW recall cannot validate omission bounds.

The official-release regression supplies a real failure mechanism in a small synthetic fixture, while SKIP and stale routing are deliberate interventions. Their complementary roles support the causal explanation of retained state but do not establish production incidence. The study assumes trusted acquisition and startup; it does not test concurrent unrecorded writers, hidden replicas or an operator returning a compliant audit copy. Hospital deployment and generated-answer risk remain unevaluated.

e) *Reproducibility and evidence provenance:* The local artifact retains paired histories, native responses, saved vectors, acquisition protocols and independent checks. The figures in this revision are derived from those saved measurements, with source hashes and counting units recorded alongside the plots. Superseded tabular presentations and adverse results remain archived; they are not additional experiments. Earlier release packages preserve their original manuscript snapshots. No public repository or archival identifier is yet available, and saved consistency alone cannot authenticate the original acquisition.

VII. CONCLUSION

Metadata exclusion is a relational claim about served state when the obligation concerns a field’s effect on retrieval scores. The evidence needed to detect a retained influence can cease to be informative after that influence is removed. Our controlled measurements explain this behavior through update propagation, reader selection and post-repair target visibility. Valid score intervals distinguish violations from insufficient evidence, while stronger bound channels can establish the finite repaired obligation at different acquisition costs. Their benefit depends on available information: complete same-state target scores remain sufficient, and high approximate-search recall supplies no substitute for a valid omission

bound. These findings motivate separate evidence plans for fault detection and repair acceptance.

REFERENCES

- [1] P. Lewis, E. Perez, A. Piktus *et al.*, “Retrieval-augmented generation for knowledge-intensive NLP tasks,” in *Advances in Neural Information Processing Systems*, vol. 33. Curran Associates, Inc., 2020, pp. 9459–9474.
- [2] J. Chen, S. Xiao, P. Zhang, K. Luo, D. Lian, and Z. Liu, “M3-embedding: Multi-linguality, multi-functionality, multi-granularity text embeddings through self-knowledge distillation,” in *Findings of the Association for Computational Linguistics: ACL 2024*. Association for Computational Linguistics, Aug. 2024, pp. 2318–2335.
- [3] S. Zeng, J. Zhang, P. He *et al.*, “The good and the bad: Exploring privacy issues in retrieval-augmented generation (RAG),” in *Findings of the Association for Computational Linguistics: ACL 2024*. Bangkok, Thailand: Association for Computational Linguistics, Aug. 2024, pp. 4505–4524.
- [4] M. Liu, S. Zhang, and C. Long, “Mask-based membership inference attacks for retrieval-augmented generation,” in *Proceedings of the ACM on Web Conference 2025*. Association for Computing Machinery, Apr. 2025, pp. 2894–2907.
- [5] A. Naseh, Y. Peng, A. Suri, H. Chaudhari, A. Oprea, and A. Houmansadr, “Riddle Me This! Stealthy Membership Inference for Retrieval-Augmented Generation,” in *Proceedings of the 2025 ACM SIGSAC Conference on Computer and Communications Security*. Association for Computing Machinery, 2025, pp. 1245–1259.
- [6] A. Thudi, H. Jia, I. Shumailov, and N. Papernot, “On the necessity of auditable algorithmic definitions for machine unlearning,” in *31st USENIX Security Symposium (USENIX Security 22)*. USENIX Association, Aug. 2022, pp. 4007–4022.
- [7] L. Xue, S. Hu, W. Lu *et al.*, “Towards reliable forgetting: A survey on machine unlearning verification,” *ACM Computing Surveys*, vol. 58, no. 12, pp. 1–35, 2026, art. no. 314.
- [8] J. Weng, S. Yao, Y. Du, J. Huang, J. Weng, and C. Wang, “Proof of unlearning: Definitions and instantiation,” *IEEE Transactions on Information Forensics and Security*, vol. 19, pp. 3309–3323, 2024.
- [9] T. Eisenhofer, D. Riepel, V. Chandrasekaran, E. Ghosh, O. Ohri-menko, and N. Papernot, “Verifiable and provably secure machine unlearning,” in *2025 IEEE Conference on Secure and Trustworthy Machine Learning (SaTML)*. IEEE, 2025, pp. 479–496.
- [10] J. Hou, A. Finke, and G. Cosma, “Neural corrective machine unranking,” *Information Sciences*, vol. 745, Jul. 2026, art. no. 123366.
- [11] S. Wang, T. Zhu, D. Ye, and W. Zhou, “When machine unlearning meets retrieval-augmented generation (RAG): Keep secret or forget knowledge?” *IEEE Transactions on Dependable and Secure Computing*, vol. 23, no. 2, pp. 2030–2045, Mar. 2026.
- [12] Z. Zeng, J. Liu, M.-F. Chiang, J. He, and Z. Zhang, “S-RAG: A novel audit framework for detecting unauthorized use of personal data in RAG systems,” in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, Jul. 2025, pp. 10 375–10 385.
- [13] Y. Li, G. Liu, C. Wang, and Y. Yang, “Generating Is Believing: Membership Inference Attacks against Retrieval-Augmented Generation,” in *ICASSP 2025 – 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2025, pp. 1–5.
- [14] Z. Qi, H. Zhang, E. P. Xing, S. Kakade, and H. Lakkaraju, “Follow my instruction and spill the beans: Scalable data extraction from retrieval-augmented generation systems,” in *The Thirteenth International Conference on Learning Representations*, 2025.

- [15] S. Zeng, J. Zhang, P. He *et al.*, “Mitigating the privacy issues in retrieval-augmented generation (RAG) via pure synthetic data,” in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, Nov. 2025, pp. 24 527–24 558.
- [16] M. Anderson, G. Amit, and A. Goldstein, “Is my data in your retrieval database? membership inference attacks against retrieval augmented generation,” in *Proceedings of the 11th International Conference on Information Systems Security and Privacy (ICISSP)*, vol. 2. SciTePress, 2025, pp. 474–485.
- [17] Y. Liu, Y. Jia, R. Geng, J. Jia, and N. Z. Gong, “Formalizing and benchmarking prompt injection attacks and defenses,” in *33rd USENIX Security Symposium (USENIX Security 24)*. USENIX Association, Aug. 2024, pp. 1831–1847.
- [18] W. Zou, R. Geng, B. Wang, and J. Jia, “PoisonedRAG: Knowledge corruption attacks to Retrieval-Augmented generation of large language models,” in *34th USENIX Security Symposium (USENIX Security 25)*. USENIX Association, Aug. 2025, pp. 3827–3844.
- [19] A. Thudi, G. Deza, V. Chandrasekaran, and N. Papernot, “Unrolling SGD: Understanding factors influencing machine unlearning,” in *2022 IEEE 7th European Symposium on Security and Privacy (EuroS&P)*. IEEE, 2022, pp. 303–319.
- [20] B. Zhang, Z. Chen, C. Shen, and J. Li, “Verification of machine unlearning is fragile,” in *Proceedings of the 41st International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 235. PMLR, Jul. 2024, pp. 58 717–58 738.
- [21] B. Zhang, Y. Dong, T. Wang, and J. Li, “Towards certified unlearning for deep neural networks,” in *Proceedings of the 41st International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 235. PMLR, Jul. 2024, pp. 58 800–58 818.
- [22] D. M. Sommer, L. Song, S. Wagh, and P. Mittal, “Athena: Probabilistic verification of machine unlearning,” *Proceedings on Privacy Enhancing Technologies*, vol. 2022, no. 3, pp. 268–290, 2022.
- [23] Y. Guo, Y. Zhao, S. Hou, C. Wang, and X. Jia, “Verifying in the dark: Verifiable machine unlearning by using invisible backdoor triggers,” *IEEE Transactions on Information Forensics and Security*, vol. 19, pp. 708–721, 2024.
- [24] C.-L. Wang, Q. Li, Z. Xiang, Y. Cao, and D. Wang, “Towards lifecycle unlearning commitment management: Measuring sample-level unlearning completeness,” in *34th USENIX Security Symposium (USENIX Security 25)*. USENIX Association, Aug. 2025, pp. 6481–6500.
- [25] S. Schelter, M. Ariannezhad, and M. de Rijke, “Forget me now: Fast and exact unlearning in neighborhood-based recommendation,” in *Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval*, ser. SIGIR ’23. Association for Computing Machinery, 2023, pp. 2011–2015.
- [26] K. Rezaei, K. Chandu, S. Feizi, Y. Choi, F. Brahman, and A. Ravichander, “RESTOR: Knowledge recovery in machine unlearning,” *Transactions on Machine Learning Research*, May 2025.
- [27] C. Chakrabortii, J. G. Alvarado, S. Abdulofizova, and S. Dwivedi, “Ghost Vectors: Soft-Deleted Embeddings Remain Reconstructible in HNSW Vector Databases,” 2026, preprint, version 1, June 16, 2026.
- [28] A. Mukkuzhi, N. Uwayisenga, Y. Bizualem, J. Kammerzell, and C. Chakrabortii, “Ghost Echoes: Semantic erasure failure in retrieval-backed applications,” arXiv:2608.20352v1, 2026, preprint.
- [29] T. Steinke, M. Nasr, and M. Jagielski, “Privacy auditing with one (1) training run,” in *Advances in Neural Information Processing Systems*, vol. 36. Curran Associates, Inc., 2023, pp. 49 268–49 280.
- [30] M. Nasr, J. Hayes, T. Steinke *et al.*, “Tight auditing of differentially private machine learning,” in *32nd USENIX Security Symposium (USENIX Security 23)*. USENIX Association, Aug. 2023, pp. 1631–1648.
- [31] G. Wang, Y. Li, Y. Liu *et al.*, “MeTMAP: Metamorphic testing for detecting false vector matching problems in LLM augmented generation,” in *Proceedings of the 2024 IEEE/ACM First International Conference on AI Foundation Models and Software Engineering*. ACM, 2024, pp. 12–23.
- [32] N. Muennighoff, N. Tazi, L. Magne, and N. Reimers, “MTEB: Massive text embedding benchmark,” in *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*. Association for Computational Linguistics, May 2023, pp. 2014–2037.
- [33] H. Su, W. Shi, J. Kasai *et al.*, “One embedder, any task: Instruction-finetuned text embeddings,” in *Findings of the Association for Computational Linguistics: ACL 2023*. Association for Computational Linguistics, Jul. 2023, pp. 1102–1121.
- [34] M. Douze, A. Guzhva, C. Deng *et al.*, “The Faiss library,” *IEEE Transactions on Big Data*, vol. 12, no. 2, pp. 346–361, Apr. 2026.
- [35] E. Schubert, “A triangle inequality for cosine similarity,” in *Similarity Search and Applications (SISAP)*. Springer, 2021, pp. 32–44.
- [36] R. Connor, L. Vadicamo, and F. Rabitti, “High-dimensional simplexes for supermetric search,” in *Similarity Search and Applications (SISAP)*. Springer, 2017, pp. 96–109.
- [37] R. Connor and L. Vadicamo, “nSimplex Zen: A novel dimensionality reduction for Euclidean and Hilbert spaces,” *ACM Transactions on Knowledge Discovery from Data*, vol. 18, no. 6, pp. 1–44, Jul. 2024, art. no. 143.
- [38] W. Nie, Z. Zhu, J. Wu, B. Luo, H. Zheng, and J.-S. R. Jang, “Identified-set geometry of distributional model extraction under Top-K censored API access,” 2026, preprint, arXiv:2605.10407v1.
- [39] LlamaIndex Contributors, “Hash of Node does not consider metadata, issue 17871,” Upstream issue report, opened February 20, 2025, accessed September 21, 2026.
- [40] —, “Include Node metadata in hash calculation, pull request 18303,” Merged March 30, commit 5884f14, 2025, accessed September 21, 2026.
- [41] —, “LlamaIndex changelog: llama-index-core 0.12.28,” Official release changelog, April 1, tag v0.12.28, 2025, records the metadata-hash fix in pull request 18303. Accessed September 21, 2026.
- [42] Y. A. Malkov and D. A. Yashunin, “Efficient and robust approximate nearest neighbor search using hierarchical navigable small world graphs,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 42, no. 4, pp. 824–836, Apr. 2020.
- [43] D. Wadden, S. Lin, K. Lo *et al.*, “Fact or fiction: Verifying scientific claims,” in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, B. Webber, T. Cohn, Y. He, and Y. Liu, Eds. Online: Association for Computational Linguistics, Nov. 2020, pp. 7534–7550.